

ThreatPulse

Operations & Analyst Reference Guide

Daily usage workflow, feature reference, prioritization & confidence scoring methodology, and a look under the hood at how data quality and reliability are engineered and validated.

PREPARED FOR

Threat Intel & SOC Analysts, Team Leads, Managers

ENRICHMENT

FIRST.org EPSS • NVD CVSS v3.1 • GreyNoise

COVERAGE

33 curated intel sources across 3 triage streams

INDICATOR FEEDS

abuse.ch (URLhaus, ThreatFox, MalwareBazaar) • OTX

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What's In This Guide

This guide is split into two halves: sections 1–3 get an analyst productive day one, and sections 4–6 explain the scoring math and the reliability engineering behind it — written for anyone who wants to understand why the data can be trusted, not just how to click through it.

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How to read this document

Every number, formula, and threshold in here was checked directly against the running source code while this guide was written, not reconstructed from memory or assumption. Where the platform will show you a live, always-current number instead (like total source count), we've said so rather than printing a figure that will drift.

SECTION 1

Platform Overview & Architecture

ThreatPulse is a high-signal cyber threat intelligence and vulnerability triage dashboard. It continuously aggregates, deduplicates, normalizes, and enriches security advisories and atomic indicators from dozens of sources into a small number of curated, prioritized views — the goal is a distraction-free surface, not a firehose.

33

CURATED SOURCES

3

TRIAGE STREAMS

4

IOC INDICATOR FEEDS

5

SOURCE TRUST TIERS

Source count as of this writing — the live, always-current figure is shown in the dashboard's Feed List panel.

The Three Triage Streams

●

Active Threat Intel & CVEs

High-urgency operational items: CISA KEV disclosures, zero-days, active RCEs, ransomware campaigns, and CVE-tagged advisories. This is where EPSS/CVSS enrichment and urgency sorting matter most.

●

Security Advisories

Official patch bulletins and platform updates — Debian, Ubuntu, AWS, FreeBSD, Kubernetes, pfSense, and similar vendor/OS security notices.

●

Informational & Research

Industry journalism, vendor research, and technical commentary that adds context without competing for triage attention (The Register, The Hacker News, SANS ISC, and similar).

🧬

IOC Watch Engine

A separate, dedicated table of atomic indicators — hashes, URLs, domains, IPs — deliberately kept out of the news streams. See Section 5 for the full confidence model.

Source Trust Tiers

Every source is classified so analysts can gauge authoritativeness at a glance — visible as a small icon next to each source throughout the dashboard, and filterable in the Feed List panel.

TIER	ICON	WHAT IT MEANS
GOV		Government or CERT source (e.g. CISA).
VENDOR		Vendor or commercial security research team.
COMMUNITY		Open-source project or community-maintained advisory feed.
OSINT		Open threat intel, including STIX/TAXII sources.
AGGREGATOR		News aggregation or independent security commentary.

How Data Flows

An automated ingestion job runs on a schedule (cron-driven), pulling every enabled source, deduplicating against both an exact-link hash and a fuzzy-title match to collapse re-reported stories, enriching CVE-tagged items with EPSS and CVSS where available, ingesting and scoring atomic indicators separately, and finally writing two static JSON payloads (news feed and IOC Watch) that the browser reads directly — there's no live backend to query, which keeps the dashboard fast and keeps the attack surface small.

SECTION 2

Analyst Daily Workflow

This is a suggested rhythm, not an enforced schedule — adapt the cadence to your team's actual shift pattern. The goal is simply to make sure the highest-signal items never sit unread.

- 1 Shift handover:** Confirm *Today's Triage Briefing (24h)* is active (it's the default scope) and check the KEV Ticker in Executive Highlights for anything new published overnight.
- 2 Urgency sort:** Switch the sort control from *Newest* to *Urgent*. Work top-down through Active Threat Intel & CVEs — this tier-then-risk-score ordering (see Section 4) surfaces the items that matter most first.
- 3 Executive reporting:** Click *Digest* to copy a formatted Markdown summary of today's top critical/high items, ready to paste into a SOC channel or briefing thread.
- 4 Indicator enrichment:** Open *IOC Watch*, filter to High confidence, and export CSV to feed perimeter firewalls, EDR, or SIEM blocklists. Default view already excludes Low-confidence noise — see Section 5.
- 5 Deep search & save:** Use **Ctrl+K** (or **Cmd+K**) to jump straight to a vendor, CVE ID, or threat actor. Star anything worth tracking — it lands in the Saved filter.

Efficiency tip

The dashboard's data auto-refreshes in the background every 5 minutes with no page reload required — it's safe to leave open on a secondary monitor all shift without a manual refresh habit or memory buildup from repeated reloads.

Repeatable views

If you find yourself rebuilding the same tag filter + search + sort combination every day, save it once under *Saved Views* (next to the tag filter controls) and reload it in one click going forward. Views are named, editable, and stored only in your own browser — see the privacy note in Section 3.

Reading the Cards

Each card in the news streams carries the source, a relative timestamp, extracted tags (CVE IDs, vendors, threat types), and — where enrichment data exists — CVSS and EPSS badges. Clicking anywhere on a card opens the original source in a new tab and marks it read automatically; the read/unread state, saved items, and all filter preferences live entirely in your browser's local storage, never on a server.

SECTION 3

Feature Reference

Every control on the dashboard, grouped by where it lives.

Header Bar

CONTROL	SHORTCUT	WHAT IT DOES
Jump (Command Palette)	Ctrl/Cmd+K	Instant fuzzy-search across every item currently loaded, by title, CVE ID, vendor, or source.
Feed List	—	Full manifest of every configured source: trust tier, and a live status dot (success / failed / disabled / no data yet) from the most recent ingestion run.
IOC Watch	—	Opens the structured indicator table — see Section 5 for the full confidence model behind it.
Quick Start / Features	—	In-app orientation modals; this document is linked from Quick Start.

Status & Highlights Panel

Collapsed by default to keep the page dense; expands to reveal two side-by-side panels:

Executive Highlights (3 tabs)

KEV Ticker cycles unread critical advisories with manual pause. **Triage Checklist** shows Critical KEV / Unread / Saved counts at a glance. **Top Priority** lists the five highest-scoring items site-wide (see Section 4).

ThreatPulse Status

Feeds Synced, Deduplicated Items, Active Threats & CVEs, Last Sync time, CVE Risk-Scoring Coverage (EPSS · CVSS), and IOC Watch coverage broken down by High/Medium/Low confidence — the same tile pattern is extended every time a new capability ships.

Search & Triage Toolbar

CONTROL	WHAT IT DOES
Search bar	Free-text search across CVEs, threats, vendors, and sources.
Triage Briefing (24h) / Full Stream	Scopes the view to the last 24 hours plus active KEVs (default), or the complete retained history.
All / Unread / ★ Saved	Quick triage filter independent of tag/search filters.
Newest / Urgent sort	Newest = publish-time descending. Urgent = severity tier, then computed risk score — see Section 4.
Read All	Marks every currently visible item as read in one action.
Digest	Copies a Markdown-formatted summary of today's top Critical/High items to the clipboard.
Sync	Manually re-fetches the latest static payload without a full page reload.

Entity & Tag Filters

CONTROL	WHAT IT DOES
Checkbox grid	Filter by OS, cloud vendor, product vendor, or threat type.
Match ANY (OR) / ALL (AND)	Toggles whether selected tags are combined inclusively or must all match.
Saved Views	Save the current tags + match mode + search + sort as a named, reusable preset; delete or reload from the dropdown any time.

Privacy note

Read state, saved items, and Saved Views are all stored exclusively in your browser's local storage — nothing is sent to or persisted on a server, and there is no account system. A "Reset Saved Data" control in the footer clears all of it instantly if needed.

IOC Watch Panel

CONTROL	WHAT IT DOES
Type chips	Filter to All / Hashes / URLs / Domains / IPs.
Show low-confidence	Off by default — Low-confidence indicators are deliberately hidden so the default view stays curated rather than a raw firehose.
Sortable columns	Click any column header (Type, Value, Family, Confidence, First Seen, Sources) to re-sort.
Copy value	One-click copy of the raw indicator value for pasting into another tool.
Export CSV	Exports the currently filtered rows — including GreyNoise classification where available — ready for SIEM/EDR/firewall ingestion.
GreyNoise badge	Shown on IP rows that have been cross-referenced: a malicious, benign, or RIOT (known business service) classification. Most IPs won't show one at any given moment — see the budget note in Section 5.

Kanban Streams

CONTROL	WHAT IT DOES
Column visibility checkbox	Hide/show an entire stream without affecting its filter state.
Show More	Progressive loading — a small batch renders up front per column, more load on demand.
Mark Read / Save / Snippet	Per-card actions available on hover or via the card footer.

Command Palette Detail

Opens instantly with **Ctrl/Cmd+K** from anywhere on the page. Matches against title, CVE ID, vendor name, and threat/malware family across everything currently loaded, and jumps straight to the matching card on selection — the fastest way to answer "do we already have something on this?" mid-conversation.

SECTION 4

Prioritization & Scoring Methodology

Two related but genuinely distinct mechanisms decide what an analyst sees first. Conflating them is a common misreading of the system, so they're documented separately here exactly as implemented.

Risk Score (the shared building block)

Every CVE-tagged item that has been enriched gets a risk score blending FIRST.org's EPSS (probability of real-world exploitation) with NVD's CVSS base score (technical severity). EPSS is weighted roughly 2.3× heavier than CVSS, because "will this actually get exploited" is a sharper triage signal than "how bad would it be if it did."

RISK SCORE

$$\text{risk_score} = (\text{EPSS} \times 70) + (\text{CVSS} \times 3)$$

EPSS ranges 0–1, CVSS ranges 0–10. Items with no CVE or no enrichment yet score 0 here and fall through to recency/due-date tiebreaks instead.

Urgent Sort Mode

Used by the toolbar's Urgent sort toggle. Sorting is tier-first, not score-first:

URGENT SORT ORDER

- 1. Severity tier (Critical > High > Medium > Low > Informational)
- 2. Risk score, descending, as the tiebreak within a tier
- 3. Nearest CISA KEV remediation due date
- 4. Most recently published

An item is placed in the "critical" severity tier for a specific set of reasons — being a CISA KEV, carrying an explicit Critical severity rating from its source, or its title indicating a zero-day, active exploitation, RCE, or ransomware — not simply for mentioning a CVE ID. This keeps "Critical" a meaningful signal rather than the default state for most of the feed.

Top Priority Spotlight

Powers the 5-item "Top Priority" tab in Executive Highlights. It starts from the same risk score, then adds two more site-wide signals a per-column sort doesn't need:

PRIORITY SCORE

$$\text{priority_score} = \text{risk_score} + \text{KEV_proximity_bonus} + \text{critical_bonus}$$

COMPONENT	BEHAVIOR
KEV_proximity_bonus	0 to +30, scaling linearly as a CISA KEV remediation deadline approaches or passes — an overdue or imminent deadline adds the full +30; the bonus fades to 0 as the

Why this matters for trust

These formulas are printed here exactly as they run in production — not simplified or approximated — specifically so a manager or analyst reviewing this document can independently verify the platform isn't a black box. If you ever want to sanity-check a specific item's placement, the math above is the whole story.

SECTION 5

IOC Watch Confidence Model

Raw atomic-indicator feeds produce thousands of entries a day, and a meaningful share are noise. Every indicator gets a computed 0–100 confidence score and a High/Medium/Low label at the moment it's ingested, from up to six blended signals — five of them fully self-contained, no external API required.

1. Base Confidence by Indicator Type

INDICATOR TYPE	BASE	WHY
SHA-256 hash	75	Byte-exact match, near-zero false-positive rate.
SHA-1 hash	72	Byte-exact, slightly shorter/older hash space.
MD5 hash	68	Byte-exact, but collisions are more feasible than the SHA family.
URL	55	Path-specific, but URLs churn faster than file hashes.
Domain	40	Subject to sinkholing, takedown, and domain recycling.
IP address	25	The most volatile indicator class — shared hosting, CDNs, and IP recycling are common.

2. Corroboration Bonus

+15 points per additional independent source reporting the same indicator, **capped at +30** (i.e. the bonus maxes out once a third source corroborates — further sightings don't keep adding weight).

3. Native Source Confidence

Where a source ships its own analyst-submitted confidence (ThreatFox provides this per-indicator), the running composite score is averaged with it rather than discarded — blending our model with the original analyst's judgment instead of overriding it.

4. Freshness Decay

Decay and retention are two **separate** mechanisms, deliberately not the same number:

TYPE	DECAY STARTS	FULLY DECAYED BY	FLOOR
IP addresses	Day 3	Day 17	40% of pre-decay score
Hashes / URLs / domains	Day 10	Day 40	40% of pre-decay score

5. Allowlist Suppression

IPs inside a curated set of major CDN/cloud-provider ranges (Cloudflare, AWS, GCP, Azure, and similar) are hard-suppressed to a flat 5 / Low, eliminating the single most common source of shared-infrastructure false positives before they ever reach the table.

6. GreyNoise Cross-Reference NEW

Optional, IP-only enrichment: a malicious classification corroborates and raises the score; a benign or RIOT (known business service — cloud/CDN/scanner vendors) classification suppresses it toward Low, catching false positives the allowlist above doesn't cover. GreyNoise's free tier caps out at 50 lookups/week *total*, shared with any manual use of GreyNoise's own site, so the platform self-limits to 30/week and works through the backlog gradually rather than all at once — most IPs simply won't have a GreyNoise badge at any given moment, and that's expected, not a gap.

Retention & Cleanup

Independent of decay: any indicator not re-sighted by any feed within **7 days** is deleted outright, and each source is hard-capped at **1,500** retained indicators (oldest trimmed first) so one high-volume feed can't crowd out the rest of the table.

Confidence labels

High ≥ 70 · Medium ≥ 40 · Low < 40. The IOC Watch panel shows High + Medium by default; Low is one click away behind "Show low-confidence," never deleted from view entirely.

CVE Enrichment Pipeline

The same "corroborate, don't just trust one source" philosophy applies to the news streams' CVE enrichment:

SIGNAL	SOURCE	BEHAVIOR
EPSS score & percentile	FIRST.org (free)	Batched lookups against every CVE-tagged item still missing a score, merged in on each run.
CVSS base score & severity	NVD CVE API 2.0	Unauthenticated: ~5 requests/30s, capped at 30 lookups/run. With a confirmed NVD API key configured: 50 requests/30s, capped at 500/run — a roughly 16× throughput increase, picked up automatically the moment a key is present.

Older CVEs sometimes simply have no published CVSS data yet on NVD's side; a lookup returning nothing isn't necessarily a fault in the pipeline.

SECTION 6 · FOR MANAGERS & TEAM LEADS

Data Reliability & Validation Engineering

This section exists to answer the question a manager should ask before a team relies on any intelligence feed: how do we know this data is good, and what happens when a piece of it fails? Every mechanism below is a deliberate design decision, not an incidental side effect.

Failure Containment

Every individual source — a news feed, an indicator feed, an enrichment API — is wrapped so its failure is isolated and recorded, never fatal to the run:

- A single failing source is logged, counted, and skipped; the ingestion run continues and every other source still completes normally.
- Per-source success/failure state is tracked on every run and surfaced as a status dot in the Feed List panel — degraded sources are visible, not silently dropped.
- Sources that actively block automated requests (some government sites enforce bot-detection regardless of request headers) automatically retry through a proxy fallback before being logged as failed, recovering data that a naive single-attempt fetch would simply lose.

Rate-Limit & Budget Discipline

Every external API this platform touches has a rate limit, and every integration is built to respect it rather than risk a key getting throttled or banned:

INTEGRATION	BUDGET BEHAVIOR
NVD CVSS enrichment	Self-throttles to the applicable published rate limit (unauthenticated or keyed tier), paced between individual requests, never bursts.
GreyNoise cross-reference	Self-imposed 30/week cap against a 50/week account-wide limit, deliberately leaving headroom for manual use of the same account elsewhere.
abuse.ch & OTX indicator feeds	Fail soft: a missing or not-yet-authorized key skips only that feed with a clear log line, never blocks the rest of the run.

Tested Before It Ships

Scoring logic, budget math, and failure-recovery paths are validated with dedicated test scenarios run against the actual functions before any change reaches production — not just eyeballed. Recent examples from this build cycle:

- Confidence-score behavior verified for every signal independently — base-by-type, corroboration cap, allowlist suppression, decay curve, and the new GreyNoise adjustment — plus the interaction between them.
- A subtle bug was caught in testing where a GreyNoise classification could be silently overwritten the next time an indicator was re-sighted by another feed; fixed and re-verified before shipping, not discovered in production.
- Both the unauthenticated and API-key-enabled request tiers for NVD, and the proxy-fallback path for blocked feeds, are exercised against mocked responses covering the success, failure, and edge-case paths.

- Database schema changes are tested against a simulated copy of the existing production database before deployment, specifically to catch the class of bug where a change works on a fresh install but breaks an already-running system.

Data Hygiene & Retention

POLICY	VALUE
News item retention	30 days, automatically pruned
IOC retention	7 days without re-sighting, automatically pruned
Per-source IOC cap	1,500 indicators, oldest trimmed first
Database maintenance	Automated VACUUM every run to keep query performance stable as data turns over
Output writes	Atomic (write-to-temp, then swap) – the dashboard never reads a half-written file mid-update

No Single Point of Failure in the Scoring Model

The core IOC confidence model (base type, corroboration, native confidence, allowlist, decay) requires no external API and no paid service to function correctly on its own. NVD's API key and GreyNoise are *force-multipliers* layered on top when available — not dependencies the system falls over without. If any premium integration is unset, unconfirmed, or temporarily down, the platform degrades to its self-contained baseline rather than failing.

Source Transparency

There is no hidden or undisclosed data source. Every one of the 33 configured feeds, its trust tier, and its live health status is visible to any analyst at any time through the Feed List panel — the same manifest a manager could review to audit exactly what's feeding the dashboard.

Bottom line for managers

Data quality here isn't a claim — it's a set of specific, inspectable mechanisms: isolated failure handling, respected API budgets, a confidence model documented down to the formula, and scoring logic that gets tested against real scenarios before it ever reaches the team relying on it.

SECTION 7

Release History & Roadmap

ThreatPulse has shipped in three build phases so far, each adding a distinct layer of capability rather than expanding scope for its own sake.

PHASE 1 Enrichment Foundation

EPSS and NVD CVSS scoring wired onto every CVE-tagged item, plus a generic STIX/TAXII ingestion path proven out against a free public source.

PHASE 2 Premium Analyst UX

Top Priority spotlight, Command Palette, Saved Views, the Digest export, source trust tiers, and accurate per-feed health tracking in the Feed List panel.

PHASE 3 IOC Watch & Indicator Confidence

Structured atomic-indicator ingestion from abuse.ch and OTX, the six-signal confidence model in Section 5, and the GreyNoise cross-reference layer.

Intentionally Not Built Yet

A few upgrades were scoped and deliberately deferred rather than rushed — documented here so the team knows they're a known, tracked gap rather than an oversight:

- Broader STIX/TAXII source coverage beyond the initial proof-of-concept source.
- Org-gated intel sharing communities (ISACs, InfraGard-style feeds) — out of scope for now, since they require credential management and TLP handling this platform doesn't yet carry.

Feedback

Questions about anything in this guide, or a workflow that isn't covered? Raise it with the team lead — this document gets revised as the platform evolves, not written once and left to go stale.